

Hamza Ahmad Mir

+92 345 691 0939 | hamzamir1416@gmail.com | linkedin.com/in/hamza-mir | github.com/hamzamir1618

EDUCATION

National University of Computer and Emerging Sciences (FAST)

Bachelor of Science in Computer Science

Islamabad, Pakistan

Aug. 2022 – Apr. 2025

Relevant Coursework: Design and Analysis of Algorithms, Artificial Intelligence, Parallel & Distributed Computing, Computer Networks, Operating Systems, Database Systems, Compiler Construction, Computer Architecture

EXPERIENCE

Computer Vision Intern – KDD Labs

Islamabad, Pakistan

Jul. 2025

- Developed and tested computer vision algorithms for image preprocessing and feature extraction.
- Worked with deep learning models for visual data analysis using OpenCV and TensorFlow.
- Assisted in dataset preparation, annotation, and model evaluation.

Game Designer Intern – M-Labs

Islamabad, Pakistan

Jul. 2024 – Aug. 2024

- Designed and prototyped gameplay mechanics using Unity and C#.
- Collaborated with developers and artists to refine game feel and UI/UX; participated in playtesting and balancing.
- Gained hands-on exposure to the full game development pipeline, including scoping, iteration, and delivery.

Graphic Designer (Volunteer) – Literary, Art & Computing Societies

FAST NUCES, Islamabad

Aug. 2022 – Apr. 2025

- Built and maintained the society web portal ECHO, a JavaScript-based platform for student society management and event coordination.
- Managed digital presence and produced web-ready visual assets for society events and announcements.

PROJECTS

Systems & Networking Tools

DupCleaner | C++17, multithreading, xxHash, perceptual hashing

- Finds duplicate photos and files cluttering your storage and safely removes them, showing exactly how much space you'll free up before deleting anything.

PeerSync | C++17, TCP sockets, mDNS, OpenSSL, delta sync

- Sends files and syncs whole folders directly between two computers on the same network, with no internet or cloud storage needed, and only re-sends the parts of a file that actually changed.

Web & Full-Stack Development

ECHO | JavaScript, HTML/CSS

- A web portal that let university society members register for events and get updates, while giving admins one place to manage approvals.

RetinaVerse | Python, Deep Learning, Web

- An AI-powered web app that analyzes retinal images and flags signs of disease to support faster diagnosis.

AI & Machine Learning Applications

TumorScope | Python, Jupyter, ResNet18

- An AI tool that classifies tumor scans and highlights exactly which part of the image drove its decision, so doctors can double-check its reasoning.

EmoTrack | Python, Jupyter, OpenCV

- Reads a person's facial expression from a live webcam feed and labels their emotion in real time.

Multi-Robot A* Pathfinding / *C++*

- Plans collision-free routes for multiple robots moving through the same space at the same time.

Game Development

Re:Sling / *Unity, C#*

- A quick, physics-based mobile game prototype built in a single game jam.

Tetris & Space Shooter / *C/C++, SFML*

- Two classic arcade games built entirely from scratch, including all the game logic and rendering.

Pacman (Assembly + Parallel) / *Assembly/MASM32, C/C++*

- A full version of Pacman built at the lowest level a computer understands, plus a version where ghost logic runs simultaneously using multithreading.

SKILLS

Languages: JavaScript, Python, C/C++, C#, Java, HTML/CSS, Assembly (MASM32)

Web: React, Node.js, HTML/CSS, REST APIs, Flask

UI/UX & Design Collaboration: Interface Design, Typography, Visual Identity, Design Thinking

Frameworks & Tools: Git, Unity, OpenCV, TensorFlow, Jupyter Notebook, SFML

Databases: SQL, MongoDB

AI / ML: ResNet, Grad-CAM, CNN, deep learning pipelines

Parallel Computing: OpenMPI, OpenMP, SIMD (AVX/NEON), OpenCL

Soft Skills: Communication, Project Management, Presentation, Adaptability, Flexibility, Critical Thinking

Languages (Spoken): English (Very Good), Urdu (Native), Punjabi (Native)

LICENSES & CERTIFICATIONS

Software Engineer Certificate, *HackerRank, Jun. 2026*

- Validated software engineering & systems skills in Python and C/C++, along with SQL.

Enterprise Design Thinking Practitioner, *IBM, Jun. 2026*

- Practitioner-level certification in IBM's Enterprise Design Thinking methodology, covering UI/UX, interface design, typography, visual identity, and cross-functional design collaboration.

Frontend Development Libraries Certification, *freeCodeCamp, Jun. 2026*

- Front-end development coursework covering React, Redux, and modern JavaScript libraries.